Multi-robot systems could help people carry out complex work in manufacturing, logistics, construction, and space operations. Realizing this potential requires both efficient coordination across large teams and ways for people without robotics expertise to direct and work alongside them.

My goal is to make these large-scale multi-robot systems practical. My current work addresses the computational cost of planning through (1) **selective coordination**, which determines when robots can be planned independently and when interactions require joint planning, and (2) **accelerated planning computation**, which uses coordination structure to organize parallel work and avoids unnecessary motion validation effort. I have built this work into a collaborative research program, setting directions for graduate teams that extend these ideas. Informed by my industry and startup experience in manufacturing, I am expanding this program in two human-facing directions: interpretable planning to help people understand and compare alternatives for robotic work, and planning for human–robot teams to accommodate human choices during shared tasks.

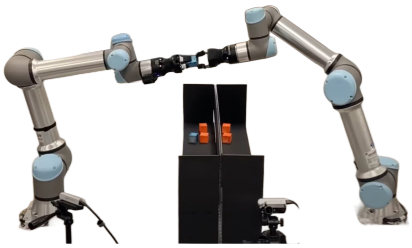


Fig. 1: Object handoff between robots.

SELECTIVE COORDINATION IN MULTI-ROBOT PLANNING

Planning each robot independently keeps the search manageable but can fail when a task requires cooperation; planning the entire system jointly captures these interactions but creates a search that grows rapidly with the numbers of robots and objects. Selective coordination makes the scope of joint reasoning a planning decision rather than a fixed global choice. I developed this idea along two complementary lines: representing coordination implied by task structure and discovering coordination revealed by motion conflicts.

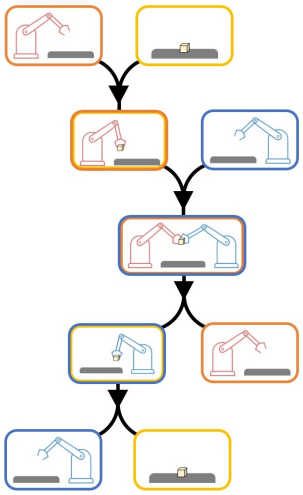


Fig. 2: Hypergraph representation of a handoff.

Representing coordination. Consider a handoff between robots: an object first moves with one robot, the two robots coordinate during the transfer, and the object then moves with the other (Figs. 1 and 2). This interaction changes which robots and objects must be considered together without requiring the entire team to remain coupled. I developed the Decomposable State Space Hypergraph (DaSH) framework, which infers these changing groups from task structure and represents their transitions without enumerating every combination across the team. For the studied rearrangement model, DaSH reduces representation growth from exponential to polynomial in the numbers of robots and objects. This change enabled planning for up to 20 objects and accelerated planning by up to 1,000× relative to the compared methods [1].

Discovering coordination. Some coordination requirements are absent from task structure. A third robot performing an independent action may cross the handoff region at the same time, creating a motion conflict. This direction grew from my work with Irving Solis, whom I mentored through completion of his Ph.D. Together, we developed Adaptive Robot Coordination (ARC) to efficiently discover and resolve such conflicts while retaining probabilistic completeness. ARC limits joint planning to affected robots and nearby motion segments, expanding the group or region as conflicts reveal the need for additional coordination [2]. Across benchmarks with varying coordination demands, ARC was the only evaluated method to find feasible motions in every scenario while maintaining competitive planning times with independent, joint, and hybrid approaches [2]. Yet a feasible motion plan may still take too long to execute. I therefore developed an anytime extension of ARC to optimize team completion time, providing almost-sure asymptotic optimality under its stated assumptions [3].

Growing the research program. DaSH and ARC grew from my earlier conflict-based approaches to multi-robot motion and task-and-motion planning [4, 5], which other research groups now use as baselines. I now set research directions for graduate teams extending these ideas to more demanding planning problems. In a student-led lazy extension of DaSH that I directed, coordination spaces are proposed and validated only when a candidate plan requires them. This enabled planning for up to 16 manipulators and 60 objects in benchmark tests, while reducing planning time by more than an order of magnitude [6]. Complementary projects use workspace layout to anticipate interactions [7, 8, 9] and explore reuse of solutions to local coordination problems [10]. Together, these projects extend selective coordination toward larger tasks with less repeated computation and provide a basis for learning coordination strategies that remain useful as tasks and team capabilities change.

ACCELERATED PLANNING INFRASTRUCTURE

Selective coordination reduces the size of the considered search space, but the remaining computation can still make planning too slow for the practical deployment of large robot teams. The structure introduced by selective coordination also creates opportunities to accelerate this work. I achieve this acceleration through two complementary lines: coordination-aware parallelism and selective validation.

Coordination-aware parallelism. Coordination structure can guide not only which robots are planned together, but also which planning computations can proceed in parallel. I led the development of a parallel optimal multi-agent pathfinding method that hierarchically combines searches for smaller robot groups, allowing independent searches to execute concurrently [11]. This work was part of my collaboration with Hannah Lee, which began during her undergraduate DREU visit and continued under my mentorship through completion of her Ph.D. at UIUC.

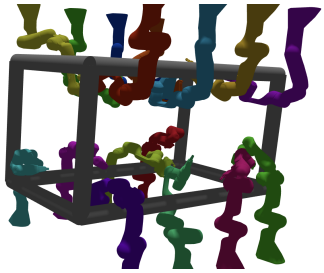


Fig. 3: Sixteen Panda robots in the cage benchmark.

In multi-robot motion planning, repairing local conflicts can change the dependencies between planning subproblems. I developed Parallel ARC to continue to exploit parallelism as these dependencies evolve, allowing independent robot searches and compatible local repairs to proceed concurrently. It was evaluated on problems with up to 256 planar robots; on a separate benchmark with 16 seven-joint robot arms (Fig. 3), it achieved a median planning time of 1.1 seconds, a 4× speedup over sequential ARC [12].

Selective validation. Collision checking becomes increasingly expensive for large robot teams because the number of robot pairs grows quadratically with team size. The amount of checking required depends on what the planner needs to decide next. To reject a candidate motion, a planner can stop after finding a single collision. To repair a team plan, it may need to locate the earliest conflict to determine which robots must coordinate and where their motions need revision. I developed reusable validation methods that tailor computation to these needs, using geometric approximations to avoid unnecessary detailed checks [13] and combining query-specific batching, early termination, and CPU vectorization (SIMD) to accelerate the rest [14]. Integrated into five planners with different coordination strategies, these methods produced end-to-end planning speedups of up to approximately 1,500× over traditional mesh-based validation implementations and brought all five methods below a one-second median runtime on a benchmark with four robot arms [14]. These accelerated planners and validation methods are available in CoMotion, an open-source library. Since its release this summer, several outside groups have adopted it, leading to new collaborations.

Selective validation also applies when the workspace changes: moving parts or equipment during reconfiguration or operation may obstruct some stored motions while leaving others unaffected. I direct student-led research that uses simplified models of the space robots occupy as they move to identify which motions remain unaffected, reusing earlier validation results for those motions and re-checking motions whose validity remains uncertain [15]. Follow-on work uses GPUs to perform many of these preliminary checks at once [16]. I will continue co-designing validation algorithms with their hardware and tailoring queries to the information needed for planning decisions. This includes evaluating proposals drawn from experience, comparing alternatives people are considering, and revising robot motions during shared work.

FUTURE DIRECTIONS

My industry collaborations and startup experience in manufacturing motivate my interest in planning systems that can adapt as tasks change and people shape how the work is done. For large robot teams, this requires anticipating new coordination needs while remaining responsive to decisions people make before and during execution. Building on selective coordination and accelerated planning, I will investigate how planners can learn and reuse coordination strategies from experience and how planning can help people direct robotic work and participate alongside robots.

Learning and reusing coordination strategies. As people direct robot teams toward more varied work, planners will encounter tasks that differ in detail but share useful coordination patterns. In assembly, experience may suggest which operations require cooperation, but a heavier component or different robot capabilities may change which robots must participate. I am interested in extending selective coordination from representing task-implied coordination and discovering motion conflicts to *predicting coordination from experience*, while retaining the ability to revise those predictions as needed.

Selective coordination also makes learning more tractable. By decomposing large planning problems into smaller interactions, it creates reusable units from which strategies can be learned and applied without considering the entire team jointly. Preliminary work on abstracting reusable strategies from task plans [17] and reusing solutions to local

motion-planning subproblems [10] provides a foundation for this approach. I aim to investigate hierarchical methods that learn both how to organize coordination across tasks and how to solve recurring local interactions, transferring useful experience without assuming that previous solutions remain valid.

Interpretable planning for human decision making. Robotic planners can optimize specified objectives, but decisions that shape robotic work often depend on human preferences that are difficult to express or become clear only after considering alternatives. In robotic assembly, for example, a product designer must balance manufacturability with other design priorities. A change that simplifies one operation may create coordination challenges elsewhere. Determining whether such a change is worthwhile requires more than a feasible plan: people need to understand how their choices affect the robot team's work so they can judge the tradeoffs.



Fig. 4: Planning with the ERUPT AR/VR platform.

Our AR/VR platform [18], developed with collaborators and a student team I mentored, supports workspace configuration and visualization of planned motions (Fig. 4). Ongoing user studies examine how people use it to specify tasks. Building on this work, I aim to develop interpretable planning methods that identify informative alternatives and explain their consequences for robotic work. This raises questions about which planning consequences matter to human judgment and how much detail must be computed to support responsive exploration without obscuring important tradeoffs. Selective coordination and accelerated planning provide foundations for focusing analysis on relevant interactions and evaluating alternatives efficiently as people refine their choices.

Planning for human–robot teams. People working with large robot teams should be able to make task-level choices without programming individual robots. In collaborative assembly, a worker's choice of the next operation can change which parts robots must deliver or support, which robots remain available, and whether later handoffs remain feasible. My prior work parallelizes computation but assumes a centralized planner that can assign every robot action and access the information needed to coordinate the team. Human–robot teams instead distribute task decisions between the planner and the person, so relevant choices may be communicated, inferred from observation, or temporarily unknown. Building on my representations of changing coordination groups and methods for local repair, I will develop planners that separate required cooperation from uncertain information about human intent and progress; determine when a choice can be handled through local retiming or replanning and when it requires reallocating work across the team; and identify which robot decisions can remain open until more is known. These planners will allow people to make task-level choices while robot teams adapt without disrupting the coordination needed to complete the shared task.

Selective coordination and accelerated computation provide a common foundation for these directions, focusing joint reasoning on the interactions a task requires and reducing the time needed to evaluate and revise plans. These capabilities could help manufacturing teams adapt to new products and logistics teams reorganize material-handling workflows. They could also support coordinated work across changing construction stages and robotic assembly or maintenance tasks in space. Across these settings, my aim is to make large robot teams practical for complex, evolving physical work, with accumulated experience informing their coordination and planning supporting the people who direct or work alongside them.

Collaboration opportunities and funding. My research has grown through collaborations that connect robot planning with complementary expertise and industrial needs. I served as primary author of a pending NSF robotics proposal with natural-language processing researchers and originated the concept for a second NSF proposal with AR/VR and human–robot interaction collaborators on workcell design for non-expert users. I have also written NSF SBIR proposals during my startup experience and served as primary author of eight industry proposals. I plan to pursue support through NSF programs such as Foundational Research in Robotics and Advanced Manufacturing, Design, and Operations and develop a CAREER proposal integrating my research and educational goals. My prior collaboration with manufacturing companies and contacts developed through my startup work provide a foundation for future industry partnerships. Established industry connections have already informed the workcell proposal and enabled me to secure a letter of collaboration. These partnerships will help motivate research questions and support evaluation in realistic manufacturing settings and other domains.

References

[1] J. D. Motes, T. Chen, T. Bretl, M. Morales, and N. M. Amato. "Hypergraph-Based Multi-Robot Task and Motion Planning". In: *IEEE Transactions on Robotics (T-RO)* (2023).

[2] I. Solis, J. D. Motes, M. Qin, M. Morales, and N. M. Amato. "Adaptive Robot Coordination: A Subproblem-based Approach for Hybrid Multi-Robot Motion Planning". In: *IEEE Robotics and Automation Letters (RA-L)* (2024).

[3] J. D. Motes, M. Morales, and N. M. Amato. "AO-ARC: Almost-Surely Asymptotically Optimal Multi-Robot Motion Planning with ARC". Manuscript under review. 2026.

[4] J. D. Motes, R. Sandström, H. Lee, S. Thomas, and N. M. Amato. "Multi-Robot Task and Motion Planning With Subtask Dependencies". In: *IEEE Robotics and Automation Letters (RA-L)* 5.2 (2020), pp. 3338–3345.

[5] I. Solis, J. D. Motes, R. Sandström, and N. M. Amato. "Representation-Optimal Multi-Robot Motion Planning Using Conflict-Based Search". In: *IEEE Robotics and Automation Letters (RA-L)* 6.3 (2021), pp. 4608–4615.

[6] S. Lee, J. D. Motes, I. Ngui, M. Morales, and N. M. Amato. "Lazy-DaSH: A Lazy Approach for Hypergraph-Based Multirobot Task and Motion Planning". In: *IEEE Transactions on Robotics (T-RO)* 42 (2026), pp. 3198–3214.

[7] C. McBeth, J. D. Motes, D. Uwacu, M. Morales, and N. M. Amato. "Scalable Multi-Robot Motion Planning for Congested Environments With Topological Guidance". In: *IEEE Robotics and Automation Letters (RA-L)* (2023).

[8] C. McBeth, J. D. Motes, I. Ngui, M. Morales, and N. M. Amato. "Scalable Multi-Robot Motion Planning Using Workspace Guidance-Informed Hypergraphs". In: *IEEE Robotics and Automation Letters (RA-L)* 11.4 (2026), pp. 4929–4936.

[9] I. Ngui, C. McBeth, J. D. Motes, M. Morales, and N. M. Amato. "Scalable Multi-Robot Motion Planning via Hierarchical Subproblem Expansion and Workspace Decomposition Refinement". In: *Algorithmic Foundations of Robotics XVII: Proc. of the Seventeenth World Symposium on the Algorithmic Foundations of Robotics (WAFR)*. 2026.

[10] I. Solis, J. D. Motes, J. Montalvo, M. Qin, M. Morales, and N. M. Amato. "Experience-Based Subproblem Planning for Multi-Robot Motion Planning". Manuscript under review. 2024.

[11] H. Lee, J. D. Motes, M. Morales, and N. M. Amato. "Parallel Hierarchical Composition Conflict-Based Search for Optimal Multi-Agent Pathfinding". In: *IEEE Robotics and Automation Letters (RA-L)* 6.4 (2021), pp. 7001–7008.

[12] J. D. Motes, M. Morales, and N. M. Amato. "P-ARC: Exploiting Subproblem Independence for Parallel Multi-Robot Motion Planning". In: *IEEE Robotics and Automation Letters (RA-L)* (2026).

[13] M. Markowicz, J. D. Motes, M. Morales, and N. M. Amato. "MR-SPITE: Accelerating Multi-Robot Conflict Scans via Hierarchical Swept-Volume Approximations". Manuscript under review. 2026.

[14] J. D. Motes, M. Morales, and N. M. Amato. "Multi-Robot Motions in Milliseconds: Vector-Accelerated Primitives for Sampling-Based Planning". Manuscript under review. 2026.

[15] S. Ashur et al. "SPITE: Simple Polyhedral Intersection Techniques for Motion Planning in Modified Environments". In: *Algorithmic Foundations of Robotics XVI, Volume 1.* Workshop on the Algorithmic Foundations of Robotics (WAFR) (2024). Vol. 37. Springer Proceedings in Advanced Robotics. Springer, 2026, pp. 151–168.

[16] Y. Arad, M. Markowicz, S. Ashur, J. D. Motes, M. Morales, and N. M. Amato. "Serialized Red-Green-Gray: Quicker Heuristic Validation of Edges in Dynamic Roadmap Graphs". Manuscript under review. 2026.

[17] K. Elimelech, J. D. Motes, M. Morales, N. M. Amato, M. Vardi, and L. E. Kavraki. "Encoding Reusable Multi-Robot Planning Strategies as Abstract Hypergraphs". In: *ICRA@40: 40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*. Rotterdam, The Netherlands, Sept. 23–26, 2024.

[18] I. Ngui et al. "ERUPT: An Open Toolkit for Interfacing with Robot Motion Planners in Extended Reality". In: *International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE)* (Goiânia, Brazil, Sept. 2026). 2026.